ADITYA SINGH

AI / Machine Learning Engineer
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PROFILE

Computer Science graduate in May 2026, specializing in machine learning engineering. I have hands-on experience designing and deploying deep learning and generative AI systems from scratch, covering everything from data pipelines and model training to production deployment on Linux and cloud infrastructure. My focus is on large language models, fine-tuning, and scalable ML infrastructure, with solid grounding in RLHF, Constitutional AI, and interpretability. I have real industry experience from internships at AIG and Primerica, plus a systems background from coursework in architecture and networking.

EDUCATION

B.S. Computer Science | Concentration: Systems & Networks *Graduated May 2026*
University of North Carolina at Charlotte | GPA: 3.5

TECHNICAL SKILLS

Languages: Python, Go, Java, JavaScript, C, C++, HTML/CSS, SQL
ML / AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, deep learning, computer vision, NLP, LLM fine-tuning, reinforcement learning, RLHF, RAG pipelines, generative AI, MLOps
Infrastructure: Docker, Linux (WSL2/Ubuntu), VirtualBox, GitHub Actions (CI/CD), ChromaDB, Gradio, Hugging Face Spaces, AWS, Tailscale, model deployment
Data & Tools: Git/GitHub, NumPy, FAISS, data pipelines, ETL, VS Code, KNIME, Microsoft Office Suite
Concepts: Transformer architecture, model training and evaluation, Constitutional AI, scalable ML systems, AUTOSAR, REST APIs, embedding search

PROJECTS

MARL Lunar Coordination Framework *Python | PyTorch | PettingZoo | PPO | Digital Twin* | 2026

- Trained 3 cooperative agents using multi-agent PPO with parameter sharing on PettingZoo's MPE simple_spread environment, targeting emergent self-organization and collision avoidance without centralized control.
- Modeled lunar comms constraints (packet loss, observation latency) as an environment wrapper; benchmarked coordination degradation across 0–40% loss rates and mid-mission agent failure scenarios.
- Built a predictive digital twin that logs full agent trajectories and runs kinematic lookahead to flag collision risks before they happen; benchmarked trained PPO against a rule-based stigmergy agent across 50 episodes -- PPO outperformed stigmergy by ~75% (-138 vs -241 mean reward) across normal, degraded comms, and agent failure conditions.

Apart Research *Python | LLM-as-Judge | Redwood Research | UK AISI* | March 2026

- Participated in an AI safety and control hackathon co-organized with Redwood Research and the UK AI Safety Institute, focused on evaluating model behavior under adversarial conditions.
- Built an LLM-as-judge eval framework with calibrated rubrics and inter-rater checks to test how models hold up under adversarial prompting.
- Based the eval criteria on Constitutional AI and alignment properties; outputs fed directly into safety analysis and written reporting.

MDM Match and Merge Engine *Python | OpenAI API | FAISS | Google Maps API | sentence-transformers* | 2026

- Honeywell x UNC Charlotte Capstone. Built a matching engine that finds duplicate customer records in Honeywell's MDM system, handling multilingual company names, address variations, typos, and abbreviations across a 5-level pipeline.

- Preprocessing stage handles language detection, translation, text normalization, and LLM-based abbreviation expansion before encoding records as FAISS embeddings to efficiently surface candidate pairs without brute-force comparison.
- Matching engine runs candidates through exact/fuzzy checks, geo distance verification, LLM company name and address analysis agents, and a final scoring layer that outputs a 0-100 confidence score with plain English reasoning.
- Led repo setup, CI pipeline, and dev environment configuration; debugged and integrated teammate contributions across all 3 weeks. 36 unit tests passing. Correctly classified Lockheed Martin HQ vs Arlington office as a non-match.

The Guilt Receipt *Python* | *Streamlit* | *Plotly* | *ReportLab* | *EPA / USDA / EIA data* | March 2026

- Built at the Cofoundr x SOE Hackathon. Converts a week of driving, food, water, and energy usage into real dollars, real gallons, and real hours using only government-published conversion factors from IRS, USDA, DOE, and EIA.
- Wrote all core calculation logic with full unit test coverage across driving, food, water, and energy modules.
- Built the full Streamlit frontend, a ReportLab PDF receipt generator, and a session history system with JSON import and export.

MoodMetrics *Python* | *OpenCV* | *MediaPipe* | *TensorFlow/FER* | *React* | *TypeScript* | *Vite* | *Gemini API* | 2026

- Built a real-time audience analytics platform that captures facial emotion signals and attention states from webcam feeds during live presentations, then generates a coaching report for the presenter at the end.
- Each audience client pushes timestamped emotion JSON to a central GitHub repo using a token-based upload system. The presenter triggers an API call that aggregates all client logs and processes them into a full session report.
- Fixed and integrated the backend built by teammates; built the React and TypeScript dashboard with bar, line, and donut chart visualizations surfacing exactly where audience attention dropped or emotion spiked.
- Plugged in the Gemini API to convert raw engagement data into actionable coaching feedback delivered as the final presenter report.

Mock ECU *Go 1.24* | *WSL2* | *GitHub* | 2026

- Co-built a mock Electronic Control Unit in Go that simulates how real AUTOSAR automotive software is structured, including modular runnable tasks, scheduling layers, and inter-component communication.
- Managed collaboration over GitHub with a proper branching strategy and code reviews — good exercise in both systems architecture and remote engineering workflow.

Interrogation Control Protocol *Python* | *Claude API* | *OpenRouter* | *GPT-4o* | March 2026

- Built for the Apart Research AI Control Hackathon 2026. Before an AI agent runs any bash command, a separate trusted monitor asks it two pointed questions about what it is doing and why. Based on the answers, the monitor approves or rejects the action.
- Claude Sonnet 4 as monitor caught 100% of malicious actions (11.1% false positive rate, 14/15 tasks completed). In every malicious case tested, the agent admitted its hidden side goal when asked directly — even when its system prompt told it to hide it.
- Cross-model run used GPT-4o via OpenRouter as monitor to prevent a misaligned Claude agent from exploiting blind spots in a Claude monitor. GPT-4o also achieved 100% catch rate but failed to return valid JSON on 60% of tasks, making it unusable in production.

Biosafety Prompt Robustness Evaluation *Python* | *Claude API* | *OpenAI API* | *LLM-as-Judge* | 2026

- Built for the AIxBio Hackathon 2026, organized by Apart Research, BlueDot Impact, and Cambridge Biosecurity Hub. Evaluated whether professional authority framing (claiming to be a biosecurity researcher, PhD student, or science journalist) shifts frontier LLM safety refusals on biology misuse queries.
- Tested 15 prompts (5 benign, 5 direct misuse, 5 adversarially framed) across Claude Sonnet 4.6 and GPT-4o. GPT-4o complied fully with 3 of 5 adversarial prompts; Claude complied with 1 of 5. Neither model fully refused any adversarially framed query.
- Key finding: GPT-4o is slightly stricter on direct misuse but significantly more permissive under adversarial framing — an audit using only direct queries reaches the opposite safety ranking from one using adversarial rephrasing.

EXPERIENCE

Data & IT Intern | Primerica *Summer 2025*

- Handled the digital conversion and validation of client financial records, making sure everything met compliance standards before it hit the system.
- Extracted structured data from documents and got it into CRM databases accurately, working alongside the IT team on their existing workflows.
- Set up error-checking and validation protocols that made the document processing pipeline significantly more reliable across thousands of records.

Software Engineering Intern | AIG *Summer 2019*

- Collaborated daily with the Japan-based overnight engineering team to keep enterprise insurance database systems running and improving across time zones.
- Wrote SQL queries and stored procedures that the company relied on for business reporting and analytics across multiple departments.
- Automated data validation and ETL processes in Python that had previously been handled manually, saving meaningful time for the team.
- Participated in cross-timezone standups every day and wrote up technical documentation to help the US and Japan teams stay aligned.

Server / Bartender | H2 Public House *Aug 2024 – Present*

- Took care of 200 or more guests per night in a high-volume sports bar, consistently delivering accurate, friendly service while staying composed under real pressure and keeping tight coordination with kitchen and bar staff.